Worksheet 1.2 — The Limit of a Function

Grade 12 Calculus and Vectors (MCV4U) · Unit 1: Rates of Change and Limits

The Big Ideas

$\lim_{x\to a} f(x) = L$ is the value f approaches near a. Substitute when continuous; for $\frac{0}{0}$ factor or rationalize first.

- Direct substitution when continuous.
- $\frac{0}{0}$ indeterminate $\Rightarrow$ factor / rationalize, then substitute.
- One-sided limits must agree; at infinity, divide by the highest power.

The Big Picture

What a limit is

$\lim_{x\to a} f(x) = L$ means $f(x)$ gets arbitrarily close to L as x approaches a regardless of the value (or gap) at a itself.

One-sided limits

The limit exists only when the left-hand and right-hand limits agree; a jump between them means the limit does not exist.

Indeterminate forms

A form like $\frac{0}{0}$ signals more work factor, rationalize, or simplify first, then substitute.

Worked Examples

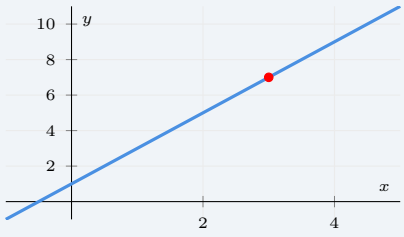
Example 1 — Direct substitution

$\lim_{x\to 3}(2x+1)$.

Solution. **Step 1 (check continuity):** $2x+1$ is a polynomial, so it's continuous everywhere — direct substitution applies.

Step 2 (substitute): $2(3)+1$.

Conclusion: $= 7$. The line passes straight through $(3, 7)$ — no hole, no jump:



Example 2 — Factor out 0/0

$\lim_{x\to 2} \dfrac{x^2-4}{x-2}$.

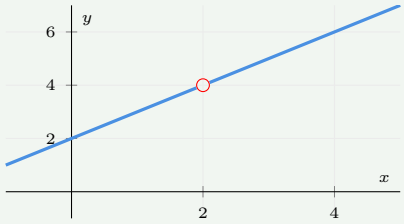
Solution. **Step 1 (try substitution):** $\dfrac{2^2-4}{2-2} = \dfrac{0}{0}$ — indeterminate, so we must factor.

Step 2 (factor the numerator): $x^2-4 = (x-2)(x+2)$.

Step 3 (cancel the common factor): $\dfrac{(x-2)(x+2)}{x-2} = x+2$ (valid for $x \neq 2$).

Step 4 (substitute into the simplified form): $2+2=4$.

Conclusion: $=4$. The graph is $y=x+2$ with a hole at $(2,4)$:



Example 3 — Rationalize 0/0

$\lim_{x\to 0} \dfrac{\sqrt{x+9}-3}{x}$.

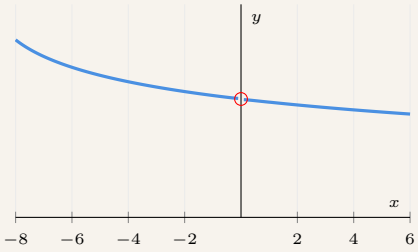
Solution. **Step 1 (try substitution):** $\dfrac{\sqrt{9}-3}{0} = \dfrac{0}{0}$ — indeterminate, so we rationalize.

Step 2 (multiply by the conjugate): $\dfrac{\sqrt{x+9}-3}{x} \cdot \dfrac{\sqrt{x+9}+3}{\sqrt{x+9}+3} = \dfrac{(x+9)-9}{x(\sqrt{x+9}+3)} = \dfrac{x}{x(\sqrt{x+9}+3)}$.

Step 3 (cancel x): $= \dfrac{1}{\sqrt{x+9}+3}$.

Step 4 (substitute): $\dfrac{1}{\sqrt{9}+3} = \dfrac{1}{3+3}$.

Conclusion: $= \dfrac{1}{6}$. The function is undefined at $x=0$ but approaches the hole at $(0, \frac{1}{6})$:



Example 4 — One-sided (DNE)

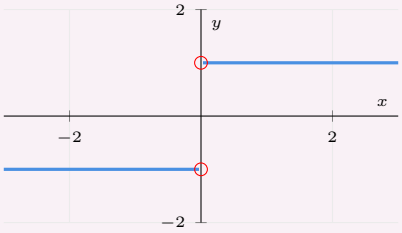
$\lim_{x\to 0} \dfrac{|x|}{x}$.

Solution. **Step 1 (right-hand limit, $x > 0$):** here $|x| = x$, so $\dfrac{|x|}{x} = \dfrac{x}{x} = 1$; thus $\lim_{x\to 0^+} = 1$.

Step 2 (left-hand limit, $x < 0$): here $|x| = -x$, so $\dfrac{|x|}{x} = \dfrac{-x}{x} = -1$; thus $\lim_{x\to 0^-} = -1$.

Step 3 (compare): $1 \neq -1$, so the one-sided limits disagree.

Conclusion: the limit does not exist. The graph jumps from -1 to 1 at $x = 0$:



Example 5 — Limit at infinity

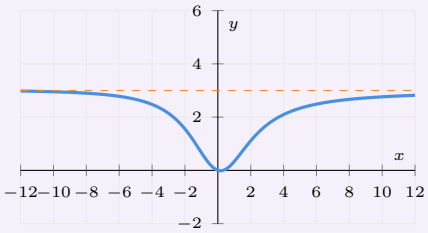
$\lim_{x\to\infty} \dfrac{3x^2 - x}{x^2 + 5}$.

Solution. **Step 1 (identify the highest power):** the highest power of x in the denominator is x^2.

Step 2 (divide every term by x^2): $\dfrac{3x^2/x^2 - x/x^2}{x^2/x^2 + 5/x^2} = \dfrac{3 - 1/x}{1 + 5/x^2}$.

Step 3 (let $x \to \infty$): $\frac{1}{x} \to 0$ and $\frac{5}{x^2} \to 0$, leaving $\dfrac{3 - 0}{1 + 0}$.

Conclusion: $= 3$ (the ratio of leading coefficients). The curve flattens toward the asymptote $y = 3$:



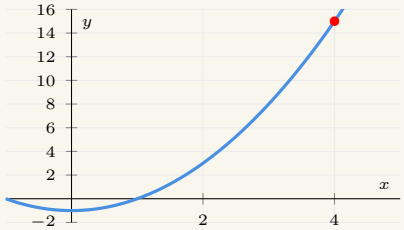
Example 6 — Direct substitution

$\lim_{x\to 4}(x^2-1)$.

Solution. **Step 1 (check continuity):** x^2-1 is a polynomial, so it's continuous everywhere.

Step 2 (substitute): $4^2-1=16-1$.

Conclusion: $=15$.



Example 7 — Factor

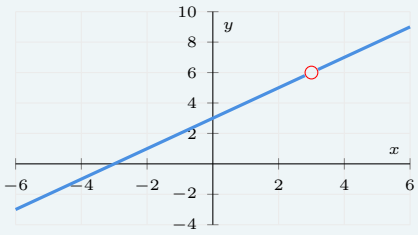
$\lim_{x\to 3}\dfrac{x^2-9}{x-3}$.

Solution. **Step 1 (try substitution):** $\dfrac{9-9}{0}=\dfrac{0}{0}$ — factor first.

Step 2 (factor and cancel): $\dfrac{(x-3)(x+3)}{x-3}=x+3$.

Step 3 (substitute): $3+3$.

Conclusion: $=6$. The graph is $y=x+3$ with a hole at $(3,6)$:



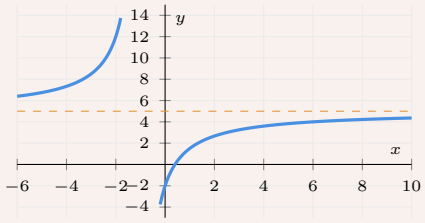
Example 8 — At infinity

$\lim_{x\to\infty} \dfrac{5x-2}{x+1}$.

Solution. **Step 1 (divide by the highest power, x):** $\dfrac{5-2/x}{1+1/x}$.

Step 2 (let $x \to \infty$): the fractions $\frac{2}{x}$ and $\frac{1}{x}$ both vanish, leaving $\dfrac{5}{1}$.

Conclusion: $= 5$ (the ratio of leading coefficients). Away from the asymptote at $x = -1$, the curve flattens toward $y = 5$:



Example 9 — Factor

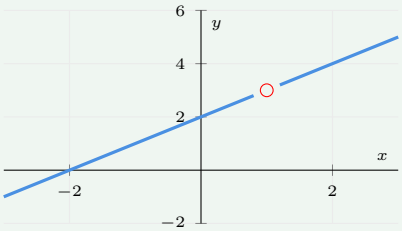
$\lim_{x\to 1} \dfrac{x^2+x-2}{x-1}$.

Solution. **Step 1 (try substitution):** $\dfrac{1+1-2}{0} = \dfrac{0}{0}$ — factor first.

Step 2 (factor the numerator): $x^2 + x - 2 = (x+2)(x-1)$.

Step 3 (cancel and substitute): $\dfrac{(x+2)(x-1)}{x-1} = x + 2 \to 1 + 2$.

Conclusion: $= 3$. The graph is $y = x + 2$ with a hole at $(1, 3)$:



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$\lim_{x \to 5}(3x - 2)$?

Question 2

$\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$?

Question 3

$\lim_{x \to \infty} \dfrac{4x - 1}{2x + 3}$?

Question 4

$\lim_{x \to 1} \dfrac{x^2 - 1}{x - 1}$?

Question 5

Does $\lim_{x \to 0} \dfrac{1}{x}$ exist?

Question 6

$\lim_{x \to -2} \dfrac{x^2 - 4}{x + 2}$?

Question 7

$\lim_{x \to \infty} \dfrac{2x^2 + 1}{x^2 - 3}$?

Question 8

$\lim_{x \to 0} \dfrac{\sqrt{x + 4} - 2}{x}$?

Question 9

$\lim_{x \to 3} (x^2 - 2x)$?

Question 10

$\lim_{x\to 4} \dfrac{x^2-16}{x-4}$?

Question 11

$\lim_{x\to \infty} \dfrac{x+5}{x^2+1}$?

Question 12

$\lim_{x\to 2} \dfrac{x^2+x-6}{x-2}$?

Question 13 — Challenge

$\lim_{x\to 0} \dfrac{\sqrt{x+1}-1}{x}$ (rationalize).

Question Recap & Answer Key

Question Recap

1. $\lim_{x\to 5}(3x-2)$?
2. $\lim_{x\to 2}\dfrac{x^2-4}{x-2}$?
3. $\lim_{x\to \infty}\dfrac{4x-1}{2x+3}$?
4. $\lim_{x\to 1}\dfrac{x^2-1}{x-1}$?
5. Does $\lim_{x\to 0}\dfrac{1}{x}$ exist?
6. $\lim_{x\to -2}\dfrac{x^2-4}{x+2}$?
7. $\lim_{x\to \infty}\dfrac{2x^2+1}{x^2-3}$?
8. $\lim_{x\to 0}\dfrac{\sqrt{x+4}-2}{x}$?
9. $\lim_{x\to 3}(x^2-2x)$?
10. $\lim_{x\to 4}\dfrac{x^2-16}{x-4}$?
11. $\lim_{x\to \infty}\dfrac{x+5}{x^2+1}$?
12. $\lim_{x\to 2}\dfrac{x^2+x-6}{x-2}$?
13. $\lim_{x\to 0}\dfrac{\sqrt{x+1}-1}{x}$ (rationalize).

✓ Answer Key

1. 13
2. 4
3. 2
4. 2
5. no ($\pm\infty$)
6. -4
7. 2
8. $\frac{1}{4}$
9. 3
10. 8
11. 0
12. 5
13. $\frac{1}{2}$